

1.5 Orthogonality and orthogonal matrices

1.5.1 Orthogonal vectors and complements

Vectors $\mathbf{u}, \mathbf{v}$ are **orthogonal**, written $\mathbf{u} \perp \mathbf{v}$, if $\mathbf{u}^\top \mathbf{v} = 0$. For orthogonal vectors the Pythagorean identity $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ holds. A set of nonzero, mutually orthogonal vectors is independent: if $\sum_j c_j \mathbf{v}_j = \mathbf{0}$, taking the inner product with $\mathbf{v}_k$ leaves $c_k \|\mathbf{v}_k\|^2 = 0$. A set is **orthonormal** if, in addition, each vector has unit length. The columns of an $n \times r$ matrix $\mathbf{Q}$ are orthonormal iff $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$.

The **orthogonal complement** of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is $\mathcal{S}^\perp = \{\mathbf{v} : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}$. The next proposition is all that this chapter needs about complements. Chapter 6 builds the theory of orthogonal projections on it.

Proposition 1.5.1

Let $\mathbf{A}$ be $m \times n$ and let $\mathcal{S} \subseteq \mathbb{R}^n$ be a subspace.

- (a) $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$ and $\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp$.
- (b) $\dim \mathcal{S}^\perp = n - \dim \mathcal{S}$, and $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp$. In particular $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ and $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$.
- (c) $(\mathcal{S}^\perp)^\perp = \mathcal{S}$.

Proof.

- (a) $\mathbf{A}\mathbf{v} = \mathbf{0}$ says that $\mathbf{v}$ is orthogonal to every row of $\mathbf{A}$, hence to their span. Apply this to $\mathbf{A}^\top$ for the second identity.
- (b) Put a basis of $\mathcal{S}$ (k vectors) into the columns of $\mathbf{B}$. Then $\mathcal{S}^\perp = \mathcal{N}(\mathbf{B}^\top)$, which has dimension $n - k$ by rank-nullity. If $\mathbf{u} \in \mathcal{S} \cap \mathcal{S}^\perp$ then $\mathbf{u}^\top \mathbf{u} = 0$, so the sum is direct and has dimension n , so it is $\mathbb{R}^n$.
- (c) $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$, and both have dimension $\dim \mathcal{S}$ by (b).

■

1.5.2 Gram–Schmidt and the QR factorization

Theorem 1.5.2 — Gram–Schmidt; QR factorization

Let $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p]$ be $n \times p$ with full column rank. Define recursively

$$\mathbf{w}_j = \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|.$$

Then $\mathbf{u}_1, \dots, \mathbf{u}_p$ are orthonormal and, for each j , span the same subspace as $\mathbf{x}_1, \dots, \mathbf{x}_j$.

Equivalently, $\mathbf{X} = \mathbf{QR}$ with $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$ satisfying $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ and $\mathbf{R}$ upper triangular with positive diagonal entries.

Proof. By induction on j . Suppose $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$ are orthonormal and span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_{j-1})$. Then $\mathbf{w}_j \neq \mathbf{0}$, since otherwise $\mathbf{x}_j$ would lie in that span, contradicting independence. For $i < j$, $\mathbf{u}_i^\top \mathbf{w}_j = \mathbf{u}_i^\top \mathbf{x}_j - \mathbf{u}_i^\top \mathbf{x}_j = 0$. So $\mathbf{u}_1, \dots, \mathbf{u}_j$ are orthonormal, and they span $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j)$ because $\mathbf{x}_j = \|\mathbf{w}_j\| \mathbf{u}_j + \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k$. That last identity, read column by column, is $\mathbf{X} = \mathbf{QR}$ with $r_{jj} = \|\mathbf{w}_j\| > 0$ and $r_{kj} = \mathbf{u}_k^\top \mathbf{x}_j$ for $k < j$. ■

The factorization is unique (Exercise (B2)). Numerical software computes it by a more stable route, but the object is the same, and Chapter 6 and Chapter 10 rely on it. Two consequences are used repeatedly. First, any orthonormal set $\mathbf{u}_1, \dots, \mathbf{u}_r$ in $\mathbb{R}^n$ extends to an orthonormal basis: apply Gram–Schmidt to $\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{e}_1, \dots, \mathbf{e}_n$, discarding each vector that depends on its predecessors. Second, every subspace has an orthonormal basis.

1.5.3 Orthogonal matrices

A square matrix $\mathbf{Q}$ is **orthogonal** if $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. By Proposition 1.3.1, $\mathbf{Q}^\top$ is then the inverse of $\mathbf{Q}$, so also $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$: the rows of an orthogonal matrix are orthonormal as well as its columns.

Proposition 1.5.3

Let $\mathbf{Q}$ and $\mathbf{P}$ be $n \times n$ orthogonal matrices.

- (a) $(\mathbf{Q}\mathbf{x})^\top (\mathbf{Q}\mathbf{v}) = \mathbf{x}^\top \mathbf{v}$ and $\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}, \mathbf{v}$.
- (b) $\mathbf{Q}^\top$ and $\mathbf{Q}\mathbf{P}$ are orthogonal.
- (c) Every entry satisfies $|q_{ij}| \leq 1$, and $\det \mathbf{Q} = \pm 1$.
- (d) If $[\mathbf{Q}_1, \mathbf{Q}_2]$ is orthogonal with $\mathbf{Q}_1$ of size $n \times r$, then $\mathbf{Q}_1 \mathbf{Q}_1^\top + \mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I}_n$, and $\mathcal{C}(\mathbf{Q}_2) = \mathcal{C}(\mathbf{Q}_1)^\perp$.

Proof.

- (a) $(\mathbf{Q}\mathbf{x})^\top (\mathbf{Q}\mathbf{v}) = \mathbf{x}^\top \mathbf{Q}^\top \mathbf{Q} \mathbf{v} = \mathbf{x}^\top \mathbf{v}$. (b) $(\mathbf{Q}\mathbf{P})^\top \mathbf{Q}\mathbf{P} = \mathbf{P}^\top \mathbf{P} = \mathbf{I}$.
- (b) Each column has unit length. For the determinant, see Proposition 1.6.2: $1 = \det(\mathbf{Q}^\top \mathbf{Q}) = (\det \mathbf{Q})^2$. (d) The first statement is $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$ written in blocks. For the second, $\mathbf{Q}_1^\top \mathbf{Q}_2 = \mathbf{0}$ gives $\mathcal{C}(\mathbf{Q}_2) \subseteq \mathcal{C}(\mathbf{Q}_1)^\perp$, and both have dimension $n - r$. ■

Part (a) says an orthogonal matrix is a rigid motion that fixes the origin: a rotation, a reflection, or a composition of the two. Changing coordinates by an orthogonal matrix changes neither lengths nor angles. This is why the distribution theory of Chapter 4 can rotate a normal vector with independent components into a convenient coordinate system without changing its distribution.

Example. (Helmert matrices)

The $n \times n$ **Helmert matrix** $\mathbf{H}$ has first row $n^{-1/2}\mathbf{1}^\top$ and, for $k = 2, \dots, n$, k th row

$$\mathbf{h}_k^\top = \frac{1}{\sqrt{k(k-1)}} \underbrace{(1, \dots, 1)}_{k-1}, -(k-1), 0, \dots, 0).$$

Each row has unit length, since $(k-1) + (k-1)^2 = k(k-1)$. Row k is orthogonal to row 1 because its entries sum to zero. For $2 \leq j < k$, the nonzero entries of $\mathbf{h}_j$ sit in positions $1, \dots, j$, where $\mathbf{h}_k$ is constant, so $\mathbf{h}_j^\top \mathbf{h}_k$ is proportional to the sum of the entries of $\mathbf{h}_j$, which is zero. So $\mathbf{H}$ is orthogonal. For $\mathbf{y} \in \mathbb{R}^n$, $(\mathbf{H}\mathbf{y})_1 = \sqrt{n}\bar{y}$ and, by Proposition 1.5.3(a),

$$\sum_{k=2}^n (\mathbf{h}_k^\top \mathbf{y})^2 = \|\mathbf{H}\mathbf{y}\|^2 - n\bar{y}^2 = \sum_{i=1}^n (y_i - \bar{y})^2.$$

The sum of squared deviations is a sum of $n-1$ squares of orthonormal contrasts. With independent normal observations this gives a short proof that the sample mean and variance are independent (Exercise (B1)).

Two other families appear later. **Permutation matrices**, the identity with its rows reordered, are orthogonal. For a unit vector $\mathbf{u}$, the **Householder reflection** $\mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$ is symmetric and orthogonal and reflects $\mathbb{R}^n$ in the hyperplane $\mathbf{u}^\perp$. Householder reflections are how QR factorizations are computed in practice (Exercise (B1)).

1.5.4 Exercises

1.5.4.1 A. Check your understanding

Exercise A1.

Show that every 2×2 orthogonal matrix is either a rotation $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ (determinant 1) or a reflection $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$ (determinant -1). Find the line fixed by the reflection.

1.5.4.2 B. Practice

Exercise B1.

Let $\mathbf{u}$ be a unit vector and $\mathbf{H} = \mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$. Show that $\mathbf{H}$ is symmetric and orthogonal, that $\mathbf{H}\mathbf{u} = -\mathbf{u}$, and that $\mathbf{H}\mathbf{v} = \mathbf{v}$ for $\mathbf{v} \perp \mathbf{u}$. Given $\mathbf{x}$ not a nonnegative multiple of $\mathbf{e}_1$, find $\mathbf{u}$ with $\mathbf{H}\mathbf{x} = \|\mathbf{x}\|\mathbf{e}_1$. Explain how repeating this column by column produces a QR factorization.

Solution. $\mathbf{H}^\top = \mathbf{H}$, and $\mathbf{H}^\top \mathbf{H} = \mathbf{I} - 4\mathbf{u}\mathbf{u}^\top + 4\mathbf{u}(\mathbf{u}^\top \mathbf{u})\mathbf{u}^\top = \mathbf{I}$. Also $\mathbf{H}\mathbf{u} = \mathbf{u} - 2\mathbf{u} = -\mathbf{u}$, and $\mathbf{H}\mathbf{v} = \mathbf{v}$ when $\mathbf{u}^\top \mathbf{v} = 0$. Let $\mathbf{w} = \mathbf{x} - \|\mathbf{x}\|\mathbf{e}_1 \neq \mathbf{0}$ and $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$. Then $\mathbf{w}^\top \mathbf{w} = 2\|\mathbf{x}\|^2 - 2\|\mathbf{x}\|x_1 = 2\mathbf{w}^\top \mathbf{x}$, so $\mathbf{H}\mathbf{x} = \mathbf{x} - 2\mathbf{w}(\mathbf{w}^\top \mathbf{x})/(\mathbf{w}^\top \mathbf{w}) = \mathbf{x} - \mathbf{w} = \|\mathbf{x}\|\mathbf{e}_1$. Applying such a reflection $\mathbf{H}_1$ to the first column of $\mathbf{X}$, then a reflection acting on coordinates $2, \dots, n$ to the second column of $\mathbf{H}_1\mathbf{X}$, and so on, gives $\mathbf{H}_p \cdots \mathbf{H}_1\mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$

with $\mathbf{R}$ upper triangular. Hence $\mathbf{X} = \mathbf{Q}\mathbf{R}$, where $\mathbf{Q}$ holds the first p columns of the orthogonal matrix $\mathbf{H}_1 \cdots \mathbf{H}_p$. In floating point one uses $\mathbf{w} = \mathbf{x} + \text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$ instead, which avoids cancellation. It gives $\mathbf{H}\mathbf{x} = -\text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$, so diagonal entries of $\mathbf{R}$ may come out negative; changing the sign of such a row of $\mathbf{R}$ and of the matching column of $\mathbf{Q}$ at the end restores the positive diagonal of [Theorem 1.5.2](#).

Exercise B2.

Show that the QR factorization of a matrix with full column rank, with $\mathbf{R}$ having positive diagonal, is unique, and that $\mathbf{R}^\top$ is the Cholesky factor of $\mathbf{X}^\top\mathbf{X}$.

Solution. If $\mathbf{X} = \mathbf{Q}_1\mathbf{R}_1 = \mathbf{Q}_2\mathbf{R}_2$, then $\mathbf{X}^\top\mathbf{X} = \mathbf{R}_1^\top\mathbf{R}_1 = \mathbf{R}_2^\top\mathbf{R}_2$. Each $\mathbf{R}_k^\top$ is lower triangular with positive diagonal, so both are the Cholesky factor of $\mathbf{X}^\top\mathbf{X}$, which is unique by Theorem 1.7.4(d). So $\mathbf{R}_1 = \mathbf{R}_2$ and $\mathbf{Q}_1 = \mathbf{X}\mathbf{R}_1^{-1} = \mathbf{Q}_2$.